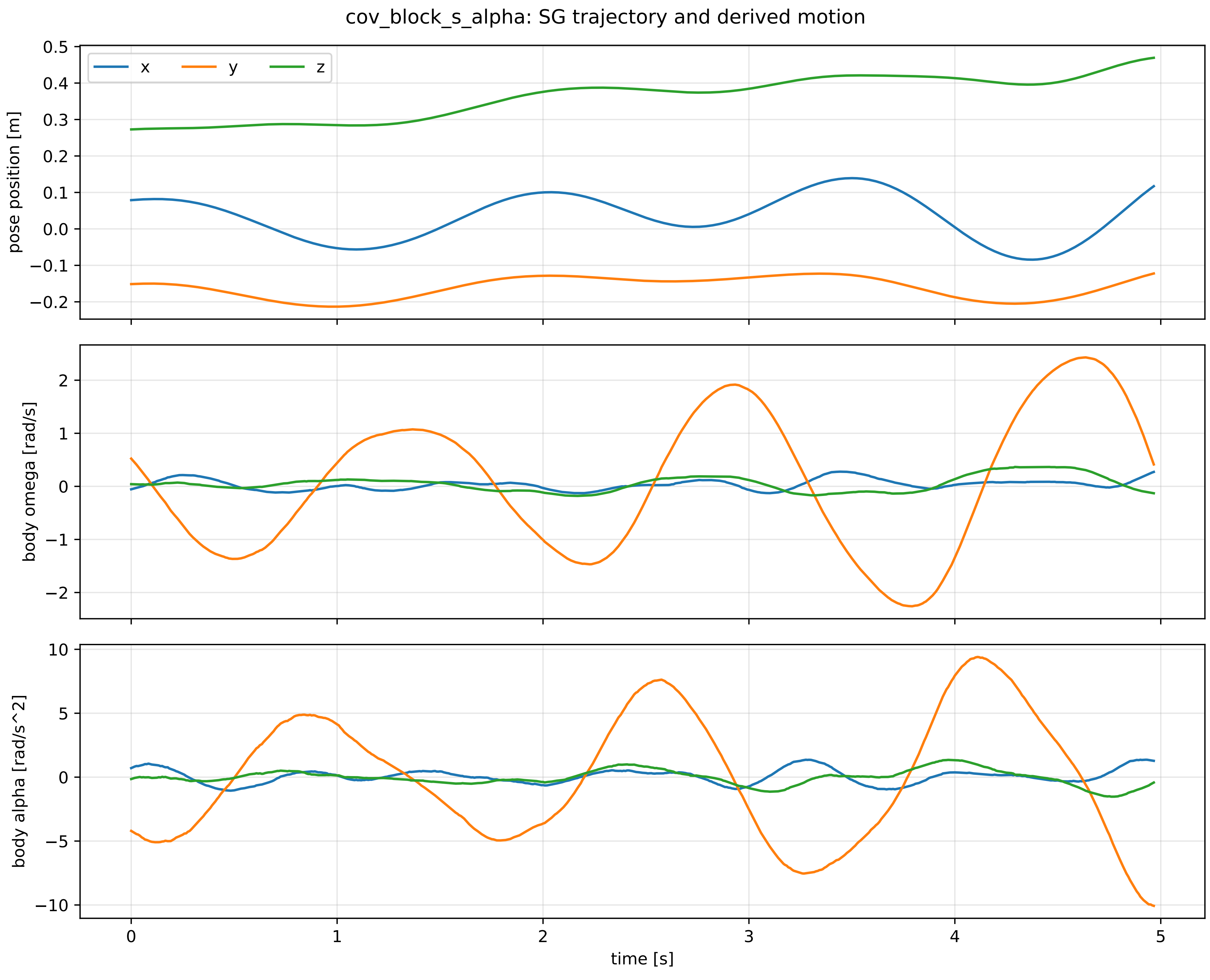
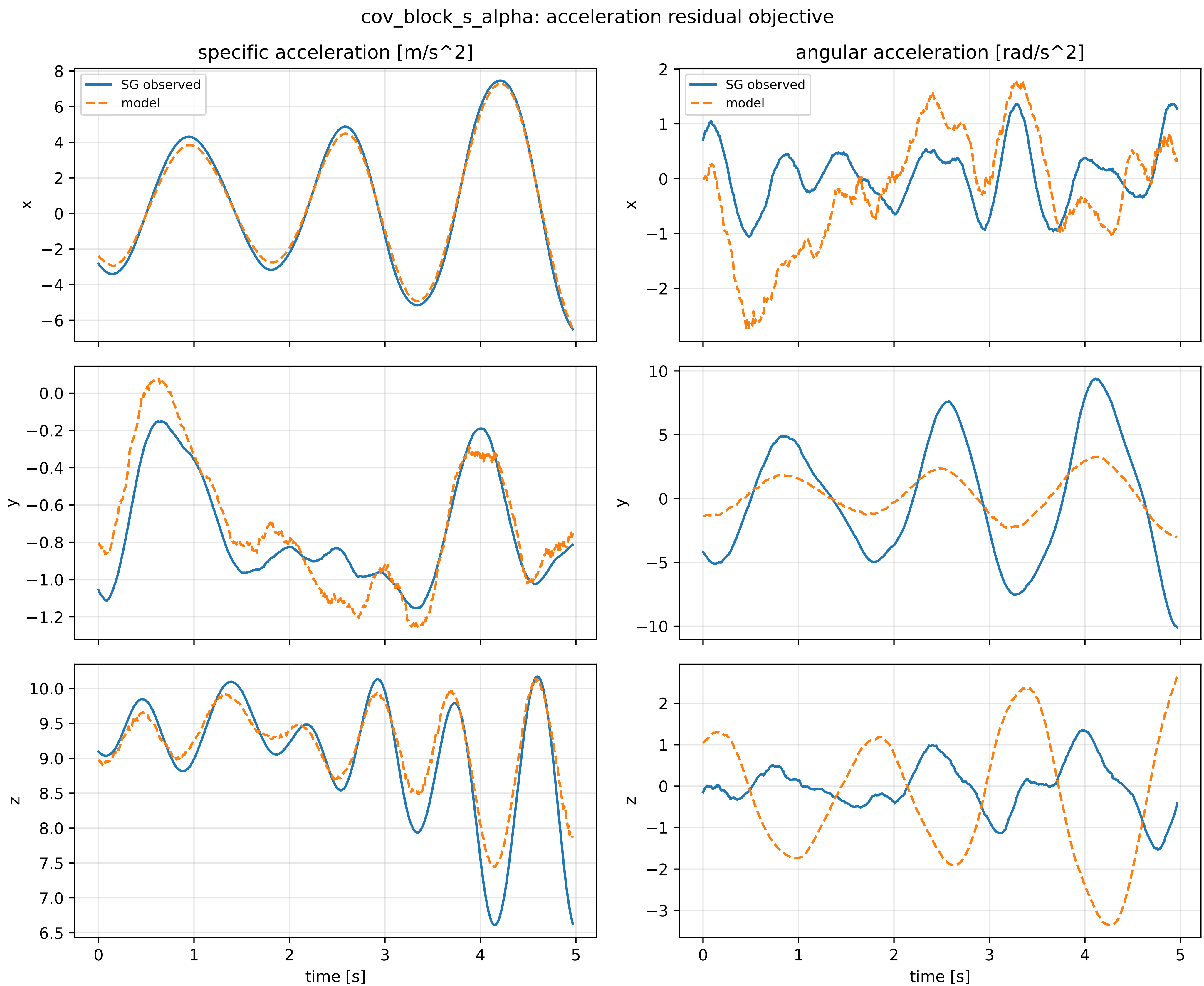
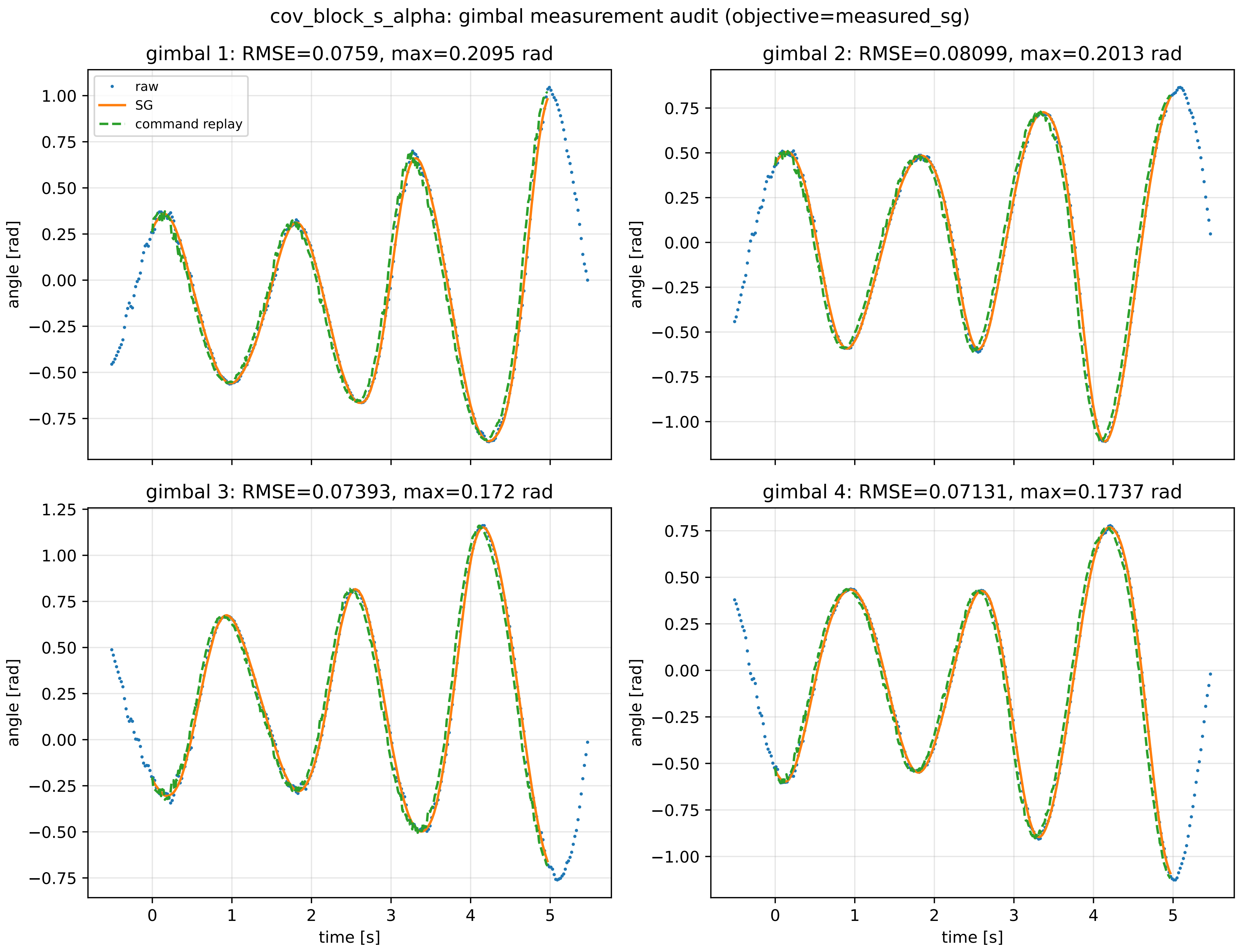


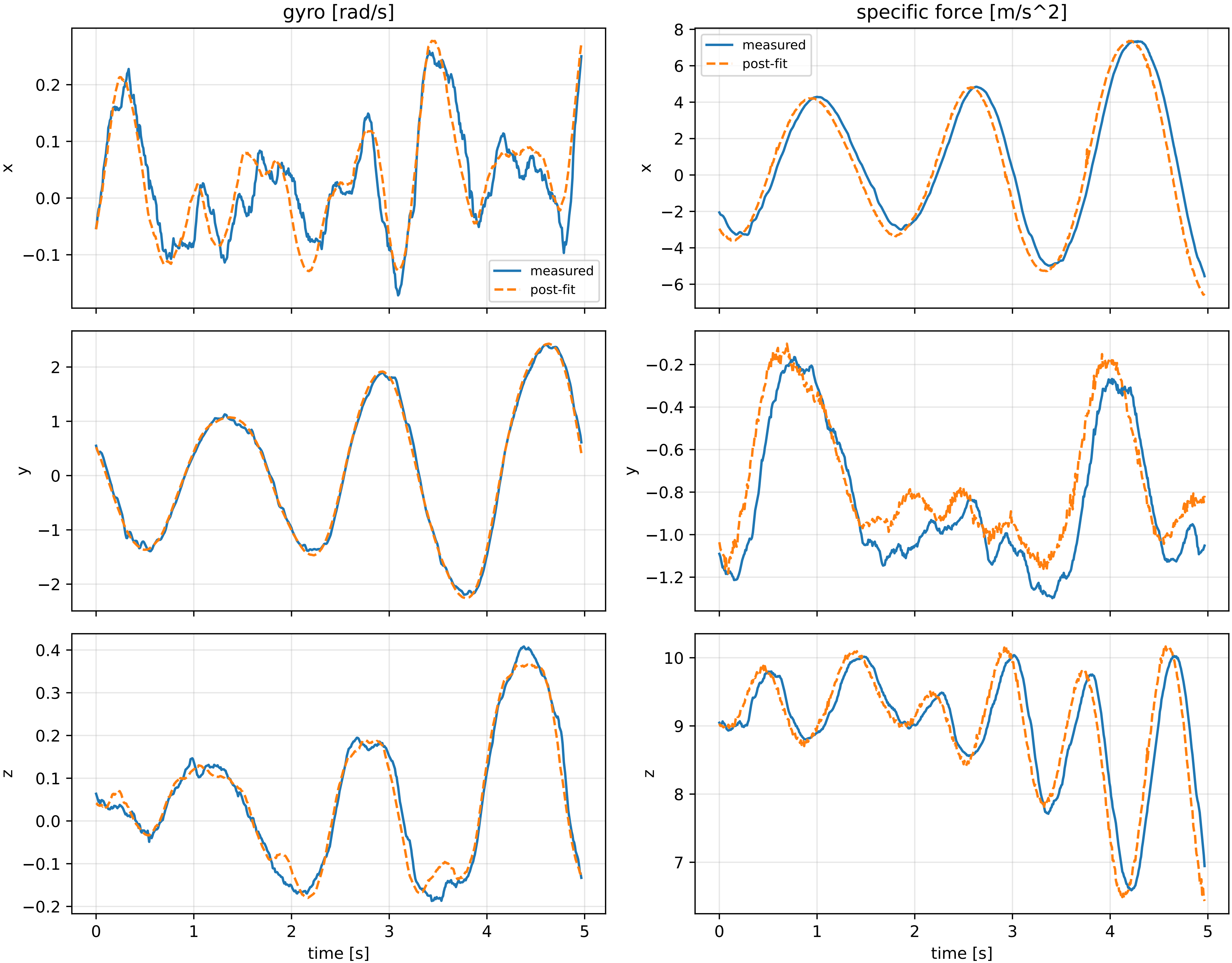
cov_block_s_alpha: SG trajectory and derived motion
pose position [m]
body omega [rad/s]
body alpha [rad/s^2]
time [s]
x
y
z
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
2
1
0
−1
−2
10
5
0
−5
−10
0
1
2
3
4
5

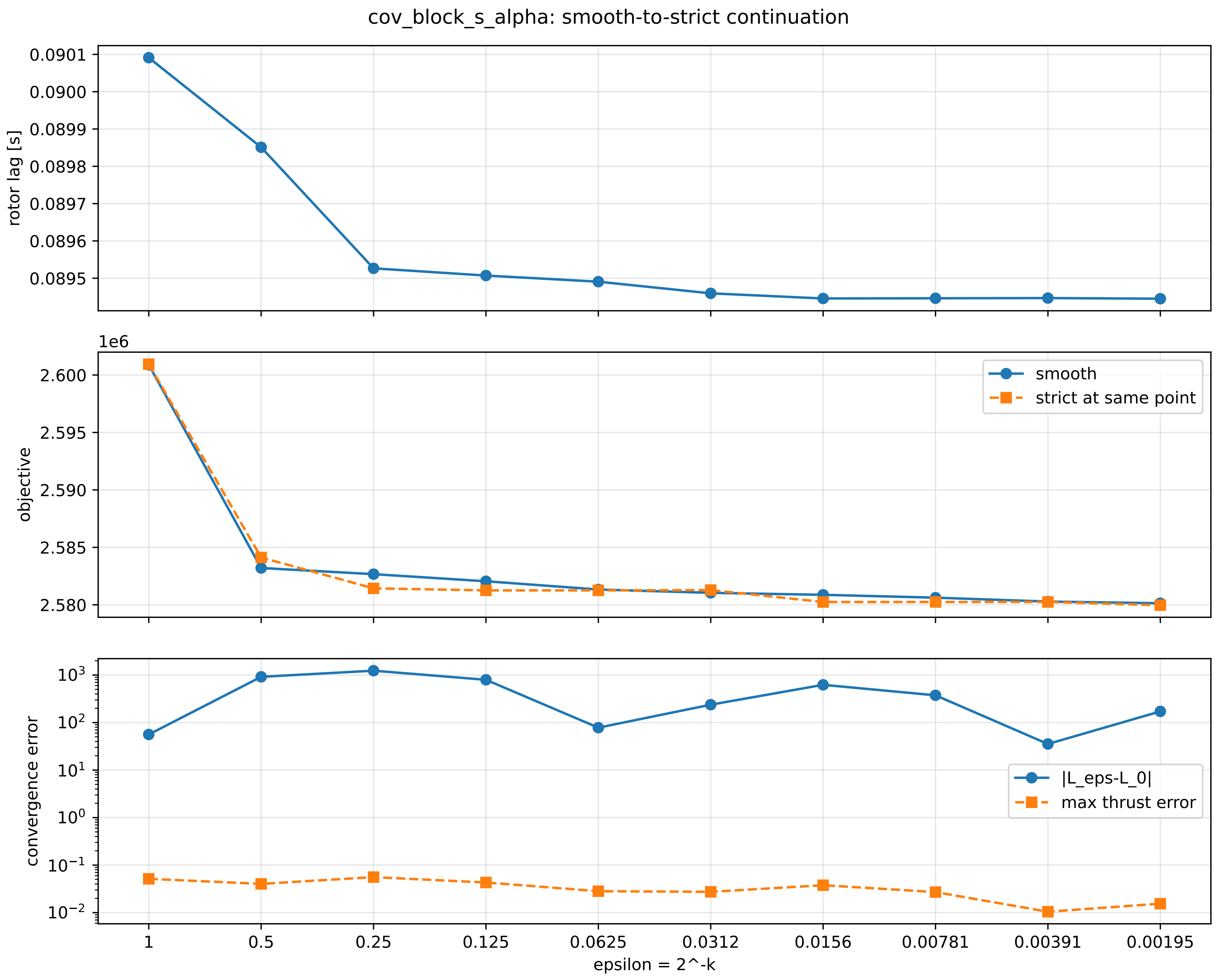

cov_block_s_alpha: acceleration residual objective
specific acceleration [m/s^2]
angular acceleration [rad/s^2]
SG observed
model
x
y
z
time [s]



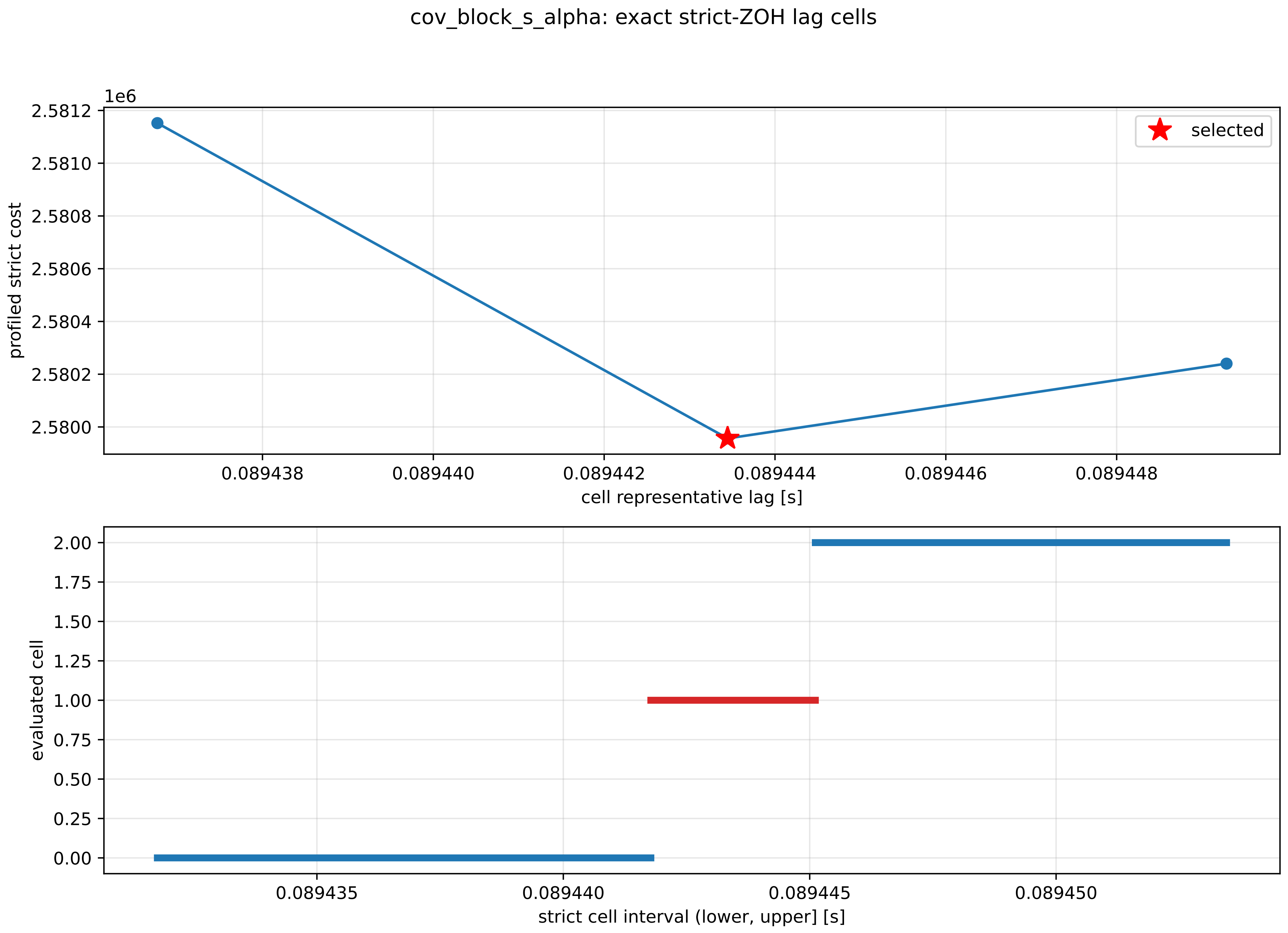
cov_block_s_alpha: gimbal measurement audit (objective=measured_sg)
gimbal 1: RMSE=0.0759, max=0.2095 rad
gimbal 2: RMSE=0.08099, max=0.2013 rad
gimbal 3: RMSE=0.07393, max=0.172 rad
gimbal 4: RMSE=0.07131, max=0.1737 rad
raw
SG
command replay
angle [rad]
time [s]

cov_block_s_alpha: IMU comparison (diagnostic only)





cov_block_s_alpha: smooth-to-strict continuation
rotor lag [s]
0.0901
0.0900
0.0899
0.0898
0.0897
0.0896
0.0895
1e6
objective
2.600
2.595
2.590
2.585
2.580
smooth
strict at same point
convergence error
10^3
10^2
10^1
10^0
10^{-1}
10^{-2}
|L_eps-L_0|
max thrust error
1
0.5
0.25
0.125
0.0625
0.0312
0.0156
0.00781
0.00391
0.00195
epsilon = 2^-k



selected (0.0894417763, 0.0894451141] s; final smooth 0.0894451125 s; data boundary=False

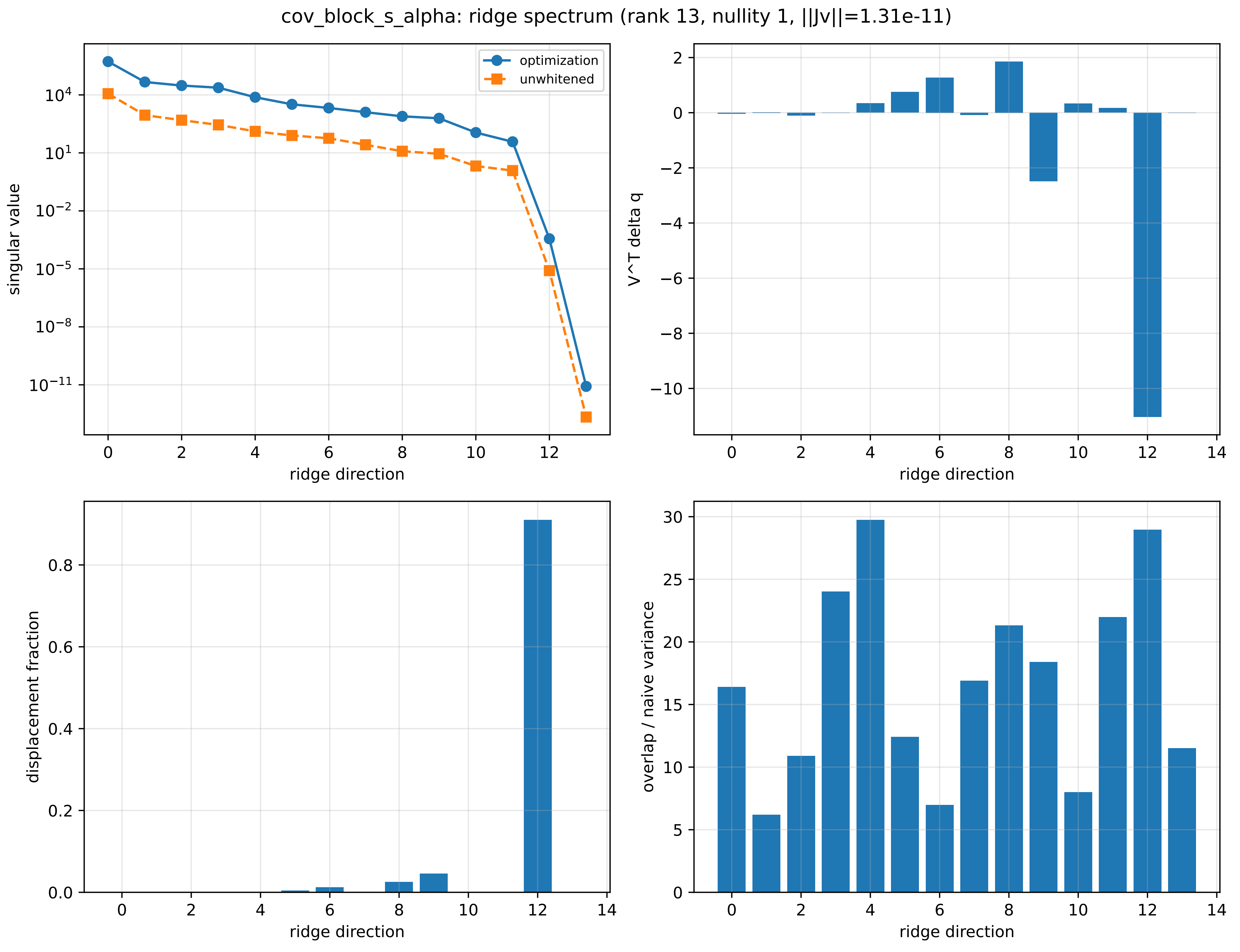
cov_block_s_alpha: parameter estimate

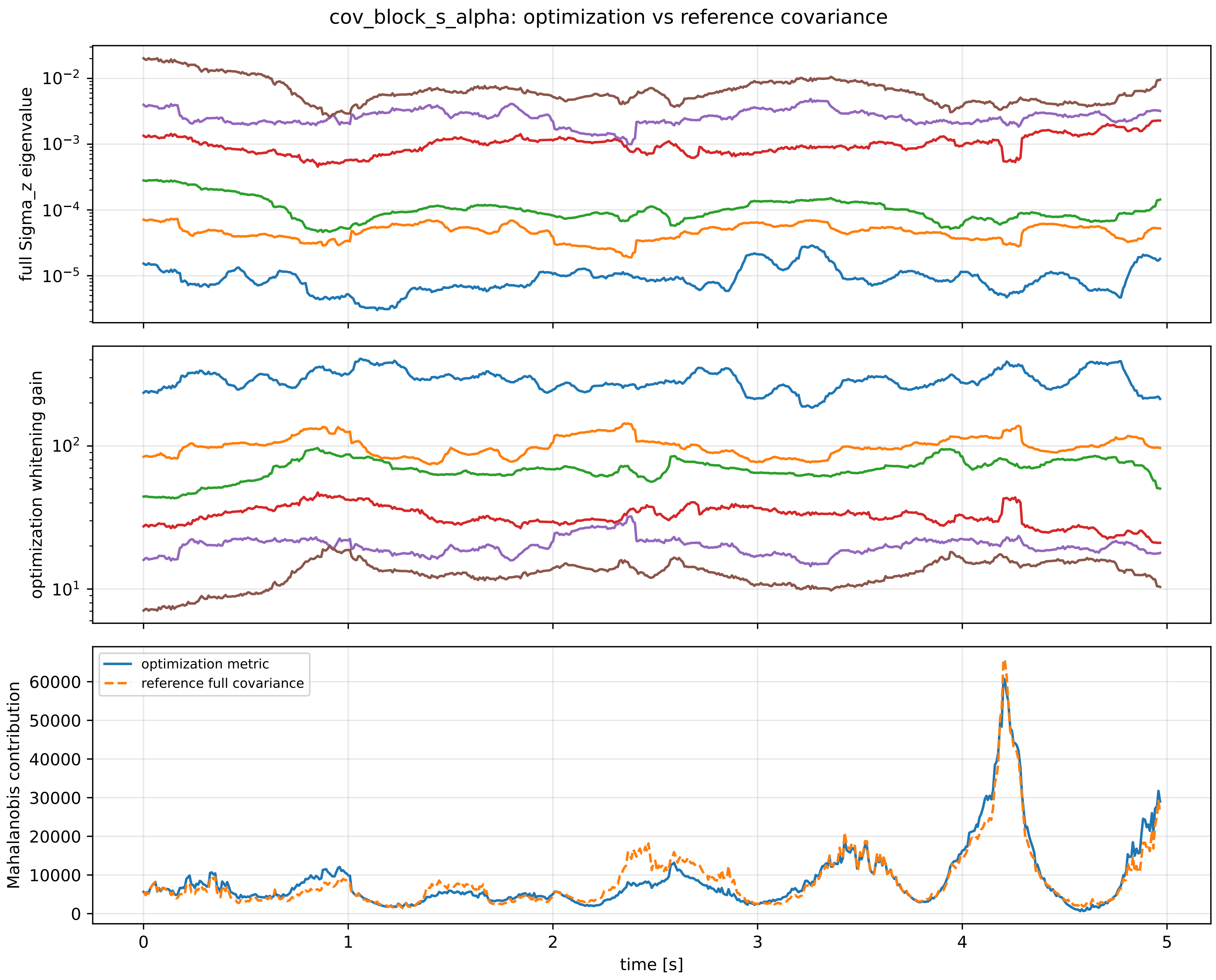
```
Case: cov_block_s_alpha
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 6.70327943
inertia [kg m^2]:
[[ 0.14732749 -0.0710711  -0.12834526]
 [-0.0710711   1.77204134 -0.04097896]
 [-0.12834526 -0.04097896  1.88005769]]
CoG body [m]: [ 0.01856588  0.13734641 -0.05461618]
force effectiveness: [0.90517497 1.20321199 4.10782951 3.61708514]

scale-free J/m [m^2]:
[[ 0.02197842 -0.01060244 -0.01914664]
 [-0.01060244  0.26435439 -0.00611327]
 [-0.01914664 -0.00611327  0.28046835]]
scale-free f/m: [0.13503465 0.17949602 0.61280893 0.53959934]

rotor lag cell: (0.0894417763, 0.0894451141] s
representative: 0.0894434452 s
```


cov_block_s_alpha: ridge spectrum (rank 13, nullity 1, ||Jv||=1.31e-11)
optimization
unwhitened
singular value
ridge direction
V^T delta q
ridge direction
displacement fraction
ridge direction
overlap / naive variance
ridge direction



cov_block_s_alpha: optimization vs reference covariance
full Sigma_z eigenvalue
10^-2
10^-3
10^-4
10^-5
optimization whitening gain
10^2
10^1
Mahalanobis contribution
optimization metric
reference full covariance
60000
50000
40000
30000
20000
10000
0
0
1
2
3
4
5
time [s]

cov_block_s_alpha: wrench / closure (force max 1.91e-14, torque max 5.33e-15)